

Technical Gates for Amplification Payment: K3–K7 Source-Ledger Burden

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Abstract

This note isolates the mathematical proof burden that is imported by the public paper *Amplification Must Be Paid*. It is a trimmed technical ledger for the downstream gates K3–K7: quantitative quotient-or-charge, forward absorption, reverse matrix absorption, high-strain transport rigidity, spectral and loop-gap certification, and compatibility with the regularity bridge.

The note starts after the upstream source-promotion data have been granted: there is a scale sequence $r_n \downarrow 0$, a candidate point z_0 , a positive source ledger $P(r_n)$, and promoted finite-thickness source families with fixed capture fraction $\eta > 0$. The purpose here is to state what K3–K7 must prove, how the constants feed the loop, and where each open analytic burden remains.

1 Scope and Inputs

This file is deliberately narrower than the full source-packing program. It does not re-prove source persistence, source promotion, or construction of the promoted family. It assumes that those upstream objects have been supplied and then asks whether the downstream ledger can close.

Assumption 1 (Upstream promoted-source package). *Along a scale sequence $r_n \downarrow 0$ at a candidate point $z_0 = (x_0, T)$, assume:*

1. $P(r_n) \geq p_* > 0$;
2. *a promoted finite-thickness source family captures a fixed fraction $\eta > 0$ of the source;*
3. *the promoted family has the usual core/collar/tail decomposition and finite-overlap bookkeeping;*
4. *quotient redundancy, if it occurs, is a genuine identification of the same physical source and not a new nonredundant amplification channel.*

The downstream proof burden is:

$$K3 + K4 + K5/K5B + K6 + K7.$$

The constant ledger at the end of the note shows why qualitative positivity is not enough: the loop closes only under a strict numerical gap.

2 Definitions and Denominator

Fix

$$Q_r(z_0) := B_r(x_0) \times (T - r^2, T), \quad \sigma_+ := (\omega \cdot S\omega)_+,$$

and define the source ledger

$$P(r) := r \iint_{Q_r(z_0)} \sigma_+ dx dt. \quad (1)$$

Assumption 2 (Local source-ledger finiteness). *For every fixed parent cylinder used in the ledger,*

$$P(r_0) = r_0 \iint_{Q_{r_0}(z_0)} \sigma_+ dx dt < \infty.$$

This is automatic in smooth pre-singular cylinders. In a suitable-weak formulation it is a standing local source-finiteness or truncation hypothesis.

The scale-critical strain variable is

$$S_r(x, t) := r^2 |S_{\text{loc}}(x, t)|,$$

and the strain row is

$$\mathfrak{B}_{\text{strain}}(r) := r \iint_{\mathcal{U}_r} \sigma_+ S_r \phi^2 dx dt = r^3 \iint_{\mathcal{U}_r} \sigma_+ |S_{\text{loc}}| \phi^2 dx dt. \quad (2)$$

The unnormalized term

$$r \iint \sigma_+ |S_{\text{loc}}| \phi^2$$

is supercritical and is not part of the live denominator.

Use the row vector

$$X(r) := \begin{pmatrix} \mathcal{E}_\omega \\ \mathcal{B}_\omega \\ \mathcal{D}_{\text{ARR}} \\ \mathfrak{B}_{\text{strain}} \\ M_{\text{deact}} \\ \mathcal{E}_\nu^{\text{coh},+} \end{pmatrix}, \quad D(r) := \mathbf{1}^T X(r). \quad (3)$$

The rows are vorticity energy-dissipation, boundary/collar flux, annular renewal/tail reservoir, source-weighted strain, deactivation, and positive coherent viscous residual.

Definition 1 (Packed boundary row). *The boundary row $\mathcal{B}_\omega(R; z_0)$ is parent-packed, not merely a single shell. It contains a one-scale shell row and the logarithmic collar packing*

$$\mathcal{B}_{\omega, \lambda_0}^{\text{pack}}(R; z_0) := \sup_{\substack{0 < a < b \\ \lambda_0 b \leq R}} \int_a^b \mathcal{B}_\omega^{\text{adv}}(s, \lambda_0; z_0) \frac{ds}{s},$$

where

$$\mathcal{B}_\omega^{\text{adv}}(s, \lambda_0; z_0) := s \iint_{Q_{\lambda_0 s}(z_0) \setminus Q_s(z_0)} |u - \bar{u}_{B_{\lambda_0 s}}(t)| |\omega|^2 |\nabla \phi_{s, \lambda_0}| \phi_{s, \lambda_0} dx dt.$$

This packed convention is part of K5 and K7 compatibility.

The mean-subtracted CKN quantity is

$$\mathcal{K}_{\text{CKN}}^{\text{ms}}(r; z_0) := r^{-2} \iint_{Q_r(z_0)} \left(|u - \bar{u}_{B_r}(t)|^3 + |p - \bar{p}_{B_r}(t)|^{3/2} \right) dx dt. \quad (4)$$

3 K3: Quantitative Quotient-or-Charge

The K3 burden is to turn nonredundant promoted packing into a positive final physical charge. The final physical ledger must include oscillation/twist, because large accumulated strain need not become net shape growth.

On each active branch core choose a source director $e_j \in S^2$ and use the sign-free projector

$$E_j := e_j \otimes e_j.$$

Define the branch oscillation density

$$\Theta_{\text{osc}}^{(j)} := \min\{1, r^2|D_t E_j| + r|\nabla E_j|\}, \quad D_t := \partial_t + u \cdot \nabla. \quad (5)$$

On an overlap $\Omega_{ij} := \mathcal{T}_i \cap \mathcal{T}_j \cap Q_r$, set

$$\Phi_{\text{osc}}(i, j) := \min \left\{ 1, \frac{r \iint_{\Omega_{ij}} \sigma_+ (|E_i - E_j| + \Theta_{\text{osc}}^{(i)} + \Theta_{\text{osc}}^{(j)}) \phi^2 dx dt}{m_{ij} + \varepsilon} \right\}, \quad (6)$$

where

$$m_{ij} := r \iint_{\Omega_{ij}} \sigma_+ \phi^2 dx dt.$$

The family oscillation ledger is the source-weighted average

$$R_{\text{osc}}^{\text{fam}}(r)P(r) := \sum_j P_j(r)R_{\text{osc}}^{(j)}(r) + o_r(1). \quad (7)$$

The final physical cost is

$$\mathcal{F}_{\text{phys}}^{\text{final}} := \mathcal{F}_{\text{phys}} + R_{\text{osc}}^{\text{fam}}. \quad (8)$$

Definition 2 (Quantitative final quotient redundancy). *Fix a quotient threshold $\delta > 0$ and let $\sim_{\delta}^{\text{final}}$ be the equivalence relation generated by significant-overlap pairs with $d_{\text{phys}}^{\text{final}} \leq \delta$, where the final edge distance is defined in (9) below. A promoted family is (δ, η) -quotient-redundant at scale r if, after finite-overlap normalization, at least $\eta P(r)$ of the captured promoted source lies inside these quotient classes and the nonclass edge mass satisfies*

$$\sum_{(i,j): i \not\sim_{\delta}^{\text{final}} j} m_{ij} < c_m^{\text{final}} \eta P(r).$$

Thus redundancy is not a verbal escape clause: it is the branch where captured mass is quantitatively accounted for by physical identification.

Proposition 1 (Oscillation ledger stabilization). *$R_{\text{osc}}^{\text{fam}}$ is a separate physical coordinate, not a shape, renewal, spread, or activity proxy. Moreover, Φ_{osc} is source-adjoint to $R_{\text{osc}}^{\text{fam}}$: a heavy nonredundant overlap with $\Phi_{\text{osc}} \geq \delta$ contributes a scale-independent positive fraction of oscillation density to the family ledger, up to finite-overlap and quotient-chain constants.*

Proof. The old shape ledger is Cauchy–Green/eigenvalue based; spread records support separation; renewal records branch identity turnover; and activity records source persistence. None of these sees $D_t E_j$, ∇E_j , or the projective direction mismatch $|E_i - E_j|$. Thus a branch may keep support, ancestry, activity, and net shape spectrum while rotating its source direction. Equation (6) integrates exactly the source-weighted density used in (7); finite-overlap normalization converts a heavy overlap into a family-level lower bound. $\square$

Required Lemma 1 (Gate K3-A: finite-overlap quotient dichotomy). *For promoted families obeying the finite-overlap bookkeeping, exactly one of the following alternatives must be available with constants independent of r :*

$$\sum_{(i,j): i \not\sim_{\delta}^{\text{final}} j} m_{ij} \geq c_m^{\text{final}} \eta P(r),$$

or the family is (δ, η) -quotient-redundant in the sense of Definition 2.

Required route. Build the finite weighted overlap graph whose vertices are promoted packets and whose edge weights are m_{ij} . Collapse the graph along $\sim_{\delta}^{\text{final}}$. Finite overlap makes total captured mass comparable to $\eta P(r)$ up to uniform constants. If the quotient-collapsed graph has too little edge mass outside its classes, the captured mass is concentrated inside quotient classes and is, by definition, the redundant branch. The contrapositive gives the displayed nonredundant mass lower bound. $\square$

Required Lemma 2 (Gate K3-B: final quantitative quotient-or-charge). *Define*

$$d_{\text{phys}}^{\text{final}}(i, j) := \max\{d_{\text{phys}}(i, j), w_{\text{osc}} \Phi_{\text{osc}}(i, j)\}. \quad (9)$$

Let $\sim_{\delta}^{\text{final}}$ be generated by significant-overlap pairs with $d_{\text{phys}}^{\text{final}} \leq \delta$, and set

$$E_{\text{nonred}}^{\text{final}} := \{(i, j) : i \not\sim_{\delta}^{\text{final}} j, m_{ij} > 0\}.$$

There are constants $c_m^{\text{final}} > 0$, $C_^{\text{final}} > 0$, and $\delta > 0$ such that, if the promoted family is not quotient-redundant and $R_{\text{pack}, \delta}^{*, \text{final}} \geq \eta$, then*

$$\sum_{(i,j) \in E_{\text{nonred}}^{\text{final}}} m_{ij} \geq c_m^{\text{final}} \eta P(r), \quad (10)$$

$$\sum_{(i,j) \in E_{\text{nonred}}^{\text{final}}} m_{ij} d_{\text{phys}}^{\text{final}}(i, j) \leq C_*^{\text{final}} P(r) \mathcal{F}_{\text{phys}}^{\text{final}}(r). \quad (11)$$

Consequently,

$$\mathcal{F}_{\text{phys}}^{\text{final}}(r) \geq c_{\text{final}}(\eta), \quad c_{\text{final}}(\eta) := \frac{\delta c_m^{\text{final}}}{C_*^{\text{final}}} \eta, \quad (12)$$

or the family is quantitatively quotient-redundant.

Required route. Every edge that survives the generated quotient satisfies $d_{\text{phys}}^{\text{final}} \geq \delta$. If the mass lower bound (10) failed, Lemma 1 routes the family into the quantitative quotient-redundant branch. The old physical coordinates give the old ledger-adjoint edge estimate, and Proposition 1 adds the oscillation estimate

$$\sum m_{ij} \Phi_{\text{osc}}(i, j) \leq C_{\text{osc}, *} P(r) R_{\text{osc}}^{\text{fam}}(r) + o_r(1).$$

Since $d_{\text{phys}}^{\text{final}} \leq d_{\text{phys}} + w_{\text{osc}} \Phi_{\text{osc}}$, the edge ledger (11) follows after enlarging C_*^{final} . The threshold $d_{\text{phys}}^{\text{final}} \geq \delta$, the mass bound, and (11) imply

$$\delta c_m^{\text{final}} \eta P(r) \leq C_*^{\text{final}} P(r) \mathcal{F}_{\text{phys}}^{\text{final}}(r),$$

and cancellation on the nontrivial source sequence gives (12). $\square$

4 K4: Forward Absorption

K4 routes the final physical charge into the denominator D .

Lemma 1 (Old forward source absorption). *There is $C_{\text{old}} < \infty$ such that*

$$P(r)\mathcal{F}_{\text{phys}}(r) \leq C_{\text{old}}D(r) + o_r(1). \quad (13)$$

Lemma 2 (Oscillation/twist forward absorption). *There is $C_{\text{osc}} < \infty$ such that*

$$P(r)R_{\text{osc}}^{\text{fam}}(r) \leq C_{\text{osc}}D(r) + o_r(1). \quad (14)$$

Proof. Using (5) and (7),

$$PR_{\text{osc}}^{\text{fam}} \lesssim r \iint_{\mathcal{U}_r} \sigma_+ (r^2 |D_t E| + r |\nabla E|) \phi^2 + o_r(1).$$

The material-direction part is controlled by the transported-director equation:

$$r^2 |D_t E| \leq CS_r + \mathcal{R}_{\text{coh}, \nu}^E.$$

It is therefore bounded by the strain row and the coherent residual row. For spatial twist, on $\{|\omega| > 0\}$,

$$|\nabla E| \leq C \frac{|\nabla \omega|}{|\omega|}.$$

Since $\sigma_+ \leq |S_{\text{loc}}||\omega|^2$, Young's inequality gives

$$r^2 \sigma_+ \frac{|\nabla \omega|}{|\omega|} \leq \frac{1}{2} r^3 \sigma_+ |S_{\text{loc}}| + Cr |\nabla \omega|^2.$$

After integration, the first term is $\mathfrak{B}_{\text{strain}}$ and the second is $\mathcal{E}_\omega$. All absorber rows are components of D . $\square$

Theorem 1 (K4-final forward absorption). *For*

$$\mathcal{F}_{\text{phys}}^{\text{final}} = \mathcal{F}_{\text{phys}} + R_{\text{osc}}^{\text{fam}},$$

there is

$$C_{\text{forw}}^{\text{final}} := C_{\text{old}} + C_{\text{osc}} \quad (15)$$

such that

$$P(r)\mathcal{F}_{\text{phys}}^{\text{final}}(r) \leq C_{\text{forw}}^{\text{final}}D(r) + o_r(1). \quad (16)$$

Together with K3,

$$P(r) \leq \frac{C_{\text{forw}}^{\text{final}}}{c_{\text{final}}(\eta)} D(r) + o_r(1). \quad (17)$$

Proof. Split $P\mathcal{F}_{\text{phys}}^{\text{final}}$ into the old physical term and $PR_{\text{osc}}^{\text{fam}}$, then apply Lemmas 1 and 2. Inequality (17) follows from (12). $\square$

5 K5B: High-Strain Transport Rigidity

K5B is the main new analytic wall. Standard deformation-gradient kinematics give forward bounds, such as shape growth controlled by accumulated strain. The needed direction is reverse: source-weighted high strain must force shape, oscillation/twist, renewal, separation, leakage, quotient redundancy, or a legitimate P -term. This is the main analytic estimate required for closure. The gates below record the required estimates and constant routes; they are the burden to prove rather than a citation-closed theorem already discharged in this note.

Lemma 3 (K5B-0 interface bridge). *Let $W \geq 0$ be any K5-admissible source-weighted integrand, including $W = S_r \mathbf{1}_{H_\Lambda}$. For promoted branches with core/collar/tail decomposition,*

$$\mathcal{T}_j = \mathcal{T}_j^{\text{core}} \cup \mathcal{T}_j^{\text{collar}} \cup \mathcal{T}_j^{\text{tail}},$$

the source-weighted integral splits as

$$r \iint_{\mathcal{U}_r} \sigma_+ W \phi^2 \leq \sum_j r \int_{E_j} \int_{I_j} \sigma_+ W(\eta_{s_j,t}^{(j)}(a), t) \phi^2 dt da + C_W(\mathcal{B}_\omega + \mathcal{D}_{\text{ARR}} + \mathcal{E}_\nu^{\text{coh},+}) + o_r(1).$$

Proof. On each core, incompressibility gives $\det D\eta_{s_j,t}^{(j)} = 1$, so the Lagrangian change of variables is exact. The complement of the core is the collar, tail, and coherent-residual set; by definition these pieces are charged to the boundary, ARR/tail, and coherent residual rows. This bridge contains no core coercivity; it only separates the interface leakage. $\square$

Required Lemma 4 (Gate K5B-A: Source-weighted core matrix rigidity). *Let F solve $\dot{F} = (\nabla u)(\eta(t), t)F$ on a branch core with $\det F = 1$, and let $E(t) = e(t) \otimes e(t)$ be the source director. For a material label a , define the source-time probability measure*

$$d\mu_{j,a}(t) := \frac{r \sigma_+(\eta_{s_j,t}^{(j)}(a), t) \phi^2(\eta_{s_j,t}^{(j)}(a), t) dt}{P_{j,a}(r) + \varepsilon}.$$

Let

$$A_{j,a}^H := \int_{\{S_r > \Lambda\}} S_r(\eta_{s_j,t}^{(j)}(a), t) d\mu_{j,a}(t).$$

Then there is C_{rig} such that

$$A_{j,a}^H \leq C_{\text{rig}}(R_{\text{shape}}^{(j,a)} + R_{\text{osc}}^{(j,a)} + 1) + o_r(1). \quad (18)$$

Required route. Partition the high-strain time set into intervals on which the projective source director varies by at most a fixed small angle. On a low-variation interval, large source-weighted dimensionless strain forces Cauchy–Green condition-number production because $\det F = 1$. Intervals not covered by this alternative are charged to projective direction variation. The finite partition cost becomes the $+1$ term; after branch averaging this is a legitimate $P(r)$ contribution. $\square$

Required Lemma 5 (Gate K5B-B: Coercive branch action). *Let*

$$A_j^H(r, \Lambda) := \frac{1}{P_j(r)} r \iint_{\{S_r > \Lambda\} \cap \mathcal{T}_j} \sigma_+ S_r \phi^2 dx dt.$$

Then

$$A_j^H \leq C_{\text{act}}(R_{\text{shape}}^{(j)} + R_{\text{osc}}^{(j)} + R_{\text{renew}}^{(j)} + R_{\text{sep}}^{(j)} + 1) + o_r(1). \quad (19)$$

Required route. Use Lemma 3 to split the branch into core and leakage. Loss of coherent identity is renewal. Persistent but physically separating branches are separation and then final physical charge. On the remaining nonrenewing, nonseparating core, Lemma 4 gives shape or oscillation/twist. The +1 term becomes $P_j(r)$ after multiplication. $\square$

Required Lemma 6 (Gate K5B-C: High-strain matrix bound). *Let $H_\Lambda = \{S_r > \Lambda\} \cap \mathcal{U}_r$. Assume K4-final and Lemma 5. If a_{leak} is the coefficient with which K5B-0 leakage rows enter D , then*

$$r \iint_{H_\Lambda} \sigma_+ S_r \phi^2 \leq a_H D(r) + C_H P(r) + o_r(1), \quad a_H := a_{\text{leak}} + C_{\text{act}} C_{\text{forw}}^{\text{final}}. \quad (20)$$

No arbitrary smallness of a_H is asserted.

Required route. Multiply (19) by $P_j(r)$, sum in j , and use that shape, oscillation, renewal, and separation are components of $\mathcal{F}_{\text{phys}}^{\text{final}}$. K4-final routes their product with $P(r)$ into $D(r)$. Leakage contributes $a_{\text{leak}} D(r)$, and the summed +1 terms contribute $C_H P(r)$. $\square$

Lemma 4 (Matrix-ready strain row). *The strain row satisfies*

$$\mathfrak{B}_{\text{strain}}(r) \leq a_{\text{strain}} D(r) + C_{\text{strain}} P(r) + o_r(1), \quad a_{\text{strain}} := a_H, \quad C_{\text{strain}} := \Lambda + C_H. \quad (21)$$

Proof. Split $\mathfrak{B}_{\text{strain}}$ into $\{S_r \leq \Lambda\}$ and H_Λ . The low-strain part is at most $\Lambda P(r)$; the high-strain part is Lemma 6. $\square$

Remark 1 (Forbidden inference). *The kinematic estimate*

$$\frac{d}{dt} \log \kappa_j(t) \leq C |S_{\text{loc}}|$$

does not imply

$$\int |S_{\text{loc}}| dt \lesssim \log \kappa_j.$$

Large accumulated strain must instead be routed to shape, oscillation/twist, renewal, separation, leakage, quotient redundancy, or P . Without the oscillation/twist channel, K5B is incomplete.

6 K5: Reverse Matrix Absorption

K5 packages all reverse row estimates into a nonnegative matrix inequality. Use the row order in (3). The required row form is:

$$\mathcal{E}_\omega \leq a_{12}\mathcal{B}_\omega + a_{13}\mathcal{D}_{\text{ARR}} + a_{14}\mathfrak{B}_{\text{strain}} + a_{16}\mathcal{E}_\nu^{\text{coh},+} + b_1P + o_r(1), \quad (22)$$

$$\mathcal{B}_\omega \leq a_{21}\mathcal{E}_\omega + a_{23}\mathcal{D}_{\text{ARR}} + a_{26}\mathcal{E}_\nu^{\text{coh},+} + b_2P + o_r(1), \quad (23)$$

$$\mathcal{D}_{\text{ARR}} \leq a_{31}\mathcal{E}_\omega + a_{32}\mathcal{B}_\omega + a_{36}\mathcal{E}_\nu^{\text{coh},+} + b_3P + o_r(1), \quad (24)$$

$$\mathfrak{B}_{\text{strain}} \leq a_{\text{strain}}\mathbf{1}^T X + C_{\text{strain}}P + o_r(1), \quad (25)$$

$$M_{\text{deact}} \leq a_{51}\mathcal{E}_\omega + a_{52}\mathcal{B}_\omega + a_{53}\mathcal{D}_{\text{ARR}} + a_{54}\mathfrak{B}_{\text{strain}} + a_{55}M_{\text{deact}} + a_{56}\mathcal{E}_\nu^{\text{coh},+} + b_5P + o_r(1), \quad (26)$$

$$\mathcal{E}_\nu^{\text{coh},+} \leq a_{61}\mathcal{E}_\omega + a_{62}\mathcal{B}_\omega + a_{63}\mathcal{D}_{\text{ARR}} + b_6P + o_r(1). \quad (27)$$

Equivalently,

$$X(r) \leq AX(r) + bP(r) + o_r(1), \quad (28)$$

where

$$A = \begin{pmatrix} 0 & a_{12} & a_{13} & a_{14} & 0 & a_{16} \\ a_{21} & 0 & a_{23} & 0 & 0 & a_{26} \\ a_{31} & a_{32} & 0 & 0 & 0 & a_{36} \\ a_{\text{strain}} & a_{\text{strain}} & a_{\text{strain}} & a_{\text{strain}} & a_{\text{strain}} & a_{\text{strain}} \\ a_{51} & a_{52} & a_{53} & a_{54} & a_{55} & a_{56} \\ a_{61} & a_{62} & a_{63} & 0 & 0 & 0 \end{pmatrix}, \quad b = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ C_{\text{strain}} \\ b_5 \\ b_6 \end{pmatrix}.$$

Theorem 2 (Matrix-ready reverse absorption). *Assume the row estimates (22)–(27) and*

$$\rho(A) < 1.$$

Then

$$D(r) \leq C_0P(r) + o_r(1), \quad C_0 := \mathbf{1}^T(I - A)^{-1}b. \quad (29)$$

Proof. Since $A \geq 0$ and $\rho(A) < 1$, the Neumann series $(I - A)^{-1} = \sum_{k \geq 0} A^k$ converges and is nonnegative. Applying it componentwise to (28) gives

$$X(r) \leq (I - A)^{-1}bP(r) + o_r(1).$$

Multiplying by $\mathbf{1}^T$ gives (29). □

For certification without explicitly inverting $I - A$, let $w_i > 0$ and set

$$q_A(w) := \max_i \frac{1}{w_i} \sum_j A_{ij}w_j.$$

Equivalently, the six weighted row sums are the row fractions of A in the weighted sup norm. Define

$$C_0^{\text{cert}}(w) := \frac{(\sum_i w_i) \max_i (b_i/w_i)}{1 - q_A(w)}. \quad (30)$$

Lemma 5 (Weighted row-sum certificate). *If $q_A(w) < 1$, then $\rho(A) \leq q_A(w) < 1$, and*

$$\mathbf{1}^T(I - A)^{-1}b \leq C_0^{\text{cert}}(w).$$

Proof. Use the weighted sup norm $\|y\|_w = \max_i |y_i|/w_i$. The operator norm of A in this norm is $q_A(w)$, so $\rho(A) \leq q_A(w)$. If $\beta_w := \max_i (b_i/w_i)$, then $(A^k b)_i \leq \beta_w q_A(w)^k w_i$. Summing over rows and $k \geq 0$ gives (30). $\square$

Remark 2 (Coefficient audit). *Every coefficient in A must come from a row proof. In particular, the strain row contributes*

$$A_{4j} = a_{\text{strain}} \quad (1 \leq j \leq 6),$$

so K5B constants directly affect the feasibility of $q_A(w) < 1$. K5 does not permit unproved smallness assumptions.

7 K6: Spectral and Loop-Gap Certification

K6 contains no PDE estimate. It is the strict algebraic loop that combines K3, K4, and K5.

Theorem 3 (Absorption loop algebra). *Assume*

$$P(r) \leq \frac{C_{\text{forw}}^{\text{final}}}{c_{\text{final}}(\eta)} D(r) + o_r(1), \quad (31)$$

$$D(r) \leq C_0 P(r) + o_r(1), \quad (32)$$

and the strict gap

$$C_{\text{forw}}^{\text{final}} C_0 < c_{\text{final}}(\eta). \quad (33)$$

Then along the contradiction sequence,

$$P(r_n) \rightarrow 0, \quad D(r_n) \rightarrow 0.$$

Proof. Let

$$q := \frac{C_{\text{forw}}^{\text{final}} C_0}{c_{\text{final}}(\eta)} < 1.$$

Combining (31) and (32) gives

$$D(r_n) \leq q D(r_n) + o_n(1),$$

and hence $D(r_n) \rightarrow 0$. Then (31) gives $P(r_n) \rightarrow 0$. $\square$

Corollary 1 (Certified loop-gap condition). *If $q_A(w) < 1$ and*

$$C_{\text{forw}}^{\text{final}} C_0^{\text{cert}}(w) < c_{\text{final}}(\eta), \quad (34)$$

then the K6 loop closes with the certified reverse constant $C_0^{\text{cert}}(w)$.

Proof. Lemma 5 supplies

$$D(r) \leq C_0^{\text{cert}}(w) P(r) + o_r(1).$$

Condition (34) is exactly the strict K6 gap using this certified reverse constant. $\square$

The single constant inequality to verify is therefore

$$C_{\text{forw}}^{\text{final}} \mathbf{1}^T (I - A)^{-1} b < \frac{\delta c_m^{\text{final}}}{C_{\text{final}}^*} \eta. \quad (35)$$

The checkable sufficient version is

$$q_A(w) < 1, \quad C_{\text{forw}}^{\text{final}} C_0^{\text{cert}}(w) < c_{\text{final}}(\eta).$$

No qualitative positivity statement can replace this strict gap.

8 K7: Compatibility and Regularity Bridge

K7 is standard regularity theory in the exact denominator normalization used by the ledger. Its burden is compatibility: every edit to D must still imply smallness of the packages used below.

Definition 3 (K7 local packages). *Let $\phi \in C_c^\infty(B_{2r})$ satisfy $\phi \equiv 1$ on B_r , and set*

$$\begin{aligned} A_\omega(2r) &:= 2r \operatorname{ess\,sup}_{t \in (T-(2r)^2, T)} \int_{B_{2r}} |\omega|^2 \phi^2 dx, \\ B_\omega^{\text{diss}}(2r) &:= \nu 2r \iint_{Q_{2r}} |\nabla \omega|^2 \phi^2 dx dt, \\ K_{\omega,3}^{\text{crit}}(r; z_0) &:= r \iint_{Q_r(z_0)} |\omega|^3 dx dt. \end{aligned}$$

The compatibility condition is

$$A_\omega(2r) + B_\omega^{\text{diss}}(2r) \leq CD(r) + o_r(1), \quad \mathcal{H}_u(r) + \mathcal{H}_p(r) + \mathcal{R}_\phi(r) \leq CD(r) + o_r(1), \quad (36)$$

where $\mathcal{H}_u, \mathcal{H}_p$ are local harmonic/projection remainders and $\mathcal{R}_\phi$ is the cutoff remainder.

Lemma 6 (Vorticity interpolation). *There is C such that*

$$K_{\omega,3}^{\text{crit}}(r; z_0) \leq C \nu^{-3/4} A_\omega(2r)^{3/4} B_\omega^{\text{diss}}(2r)^{3/4} + \mathcal{R}_\phi(r). \quad (37)$$

Proof. This is the scale-correct local Gagliardo–Nirenberg estimate applied to $\omega\phi$, integrated over a parabolic interval and multiplied by r . $\square$

Lemma 7 (Mean-subtracted velocity and pressure control). *The standard local div–curl/Biot–Savart and pressure decomposition estimates give*

$$r^{-2} \iint_{Q_r} |u - \bar{u}_{B_r}(t)|^3 \leq CK_{\omega,3}^{\text{crit}}(2r; z_0) + \mathcal{H}_u(r),$$

and

$$r^{-2} \iint_{Q_r} |p - \bar{p}_{B_r}(t)|^{3/2} \leq Cr^{-2} \iint_{Q_{2r}} |u - \bar{u}_{B_{2r}}(t)|^3 + \mathcal{H}_p(r).$$

Proof. Subtracting the ball mean fixes the local Galilean gauge. The localized Biot–Savart piece is controlled by the L^3 vorticity norm; the harmonic velocity remainder is $\mathcal{H}_u$. For pressure, write $-\Delta p_{\text{loc}} = \partial_i \partial_j [(u_i - \bar{u}_i)(u_j - \bar{u}_j)]$ on a localized ball and apply Calderon–Zygmund estimates. The harmonic pressure remainder is $\mathcal{H}_p$. $\square$

Lemma 8 (Mean-subtracted CKN epsilon criterion). *There exists $\varepsilon_{\text{CKN}} > 0$ such that*

$$\mathcal{K}_{\text{CKN}}^{\text{ms}}(r; z_0) < \varepsilon_{\text{CKN}}$$

implies that z_0 is regular.

Theorem 4 (K7 regularity bridge). *Assume (36). If $D(r_n) \rightarrow 0$ along $r_n \downarrow 0$, then*

$$\mathcal{K}_{\text{CKN}}^{\text{ms}}(r_n; z_0) \rightarrow 0.$$

Consequently z_0 is regular.

Proof. Compatibility makes the localized vorticity energy-dissipation package and all harmless remainders vanish. Lemma 6 gives $K_{\omega,3}^{\text{crit}}(2r_n; z_0) \rightarrow 0$. Lemma 7 then gives mean-subtracted velocity and pressure smallness, so $\mathcal{K}_{\text{CKN}}^{\text{ms}}(r_n; z_0) \rightarrow 0$. Apply Lemma 8. $\square$

9 Contradiction Chain

Theorem 5 (Conditional K3–K7 closure). *Assume the upstream promoted-source package, K3, K4, K5/K5B, K6, and K7 in the forms stated above. If the strict loop gap (33) holds, then the candidate point z_0 is regular.*

Proof. If the promoted family is quotient-redundant, it is not a nonredundant amplification channel. Otherwise K3 gives

$$\mathcal{F}_{\text{phys}}^{\text{final}} \geq c_{\text{final}}(\eta).$$

K4 gives

$$P \leq \frac{C_{\text{forw}}^{\text{final}}}{c_{\text{final}}(\eta)} D + o.$$

K5 gives $D \leq C_0 P + o$, with either

$$C_0 = \mathbf{1}^T (I - A)^{-1} b$$

or a certified sufficient constant $C_0^{\text{cert}}(w)$. K6 and the strict gap force $P(r_n) \rightarrow 0$ and $D(r_n) \rightarrow 0$. K7 then converts $D(r_n) \rightarrow 0$ into CKN smallness and regularity, contradicting singularity. $\square$

10 Falsification and Attack Surface

This note is useful only if it makes the failure modes visible. A critic can break the K3–K7 closure by showing any one of the following:

1. A nonredundant promoted source sequence with positive $P(r_n)$ and no final physical payment.
2. High source-weighted strain that avoids shape, oscillation/twist, renewal, separation, leakage, quotient redundancy, and a legitimate P -term.
3. A K5 matrix whose proved coefficients cannot satisfy $\rho(A) < 1$, or cannot satisfy a certified weighted row condition $q_A(w) < 1$.
4. A failure of the strict K6 loop gap

$$C_{\text{forw}}^{\text{final}} C_0 < c_{\text{final}}(\eta).$$

5. A denominator D that collapses without implying mean-subtracted CKN smallness.

These are the attack points; if one survives, the proposed closure survives only as a conditional ledger, not as a completed regularity proof.

11 Proof-Burden Ledger

Gate	Claim	Remaining burden
K3	Nonredundant promoted packing either quotient-collapses or yields $\mathcal{F}_{\text{phys}}^{\text{final}} \geq c_{\text{final}}(\eta)$.	Graph quotient constants, finite-overlap mass transfer, final physical coordinate thresholds, and the oscillation-adjoint estimate.
K4	$P\mathcal{F}_{\text{phys}}^{\text{final}} \leq C_{\text{forw}}^{\text{final}}D + o$.	Forward routing of all physical ledgers, especially $R_{\text{osc}}^{\text{fam}}$, into the chosen denominator rows with scale-independent constants.
K5B	Source-weighted high strain routes to shape, oscillation, renewal, separation, leakage, quotient redundancy, or P .	The reverse high-strain rigidity estimate (20); no substitution of an unweighted path diagnostic for the source-weighted action.
K5	The six denominator rows satisfy $X \leq AX + bP + o$, and $\rho(A) < 1$.	Row-by-row coefficient provenance, packed boundary constants, ARR/tail routing, deactivation and coherent-residual constants, and a spectral certificate such as $q_A(w) < 1$.
K6	The constants satisfy $C_{\text{forw}}^{\text{final}}C_0 < c_{\text{final}}(\eta)$.	Strict numerical loop gap; positivity of $c_{\text{final}}(\eta)$ alone is not enough.
K7	$D(r_n) \rightarrow 0$ implies mean-subtracted CKN smallness.	Compatibility of the exact denominator rows with local vorticity energy-dissipation, harmless cutoff/projection remainders, and the mean-subtracted CKN normalization.

12 Conclusion

This trimmed note is the public technical burden behind the imported gates in the amplification-payment paper. The proof is not reduced to one inequality until every row coefficient in K5 is proved, the K5B high-strain transport wall is crossed, and K6 verifies the strict constant gap. K7 is standard in spirit, but it remains coupled to the exact denominator normalization: edits to the ledger must preserve (36).